Østfold University College

Mandatory: Clustering

ITF31519-1 25H Praktisk maskinlæring

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Halden

Contents

Clustering

1.1 Clustering of Wine Quality Dataset

- What clustering algorithm is the best at finding the clusters identified by the targets (external evaluation)?

  This will be solved in the appended *Notebook*(cluster.ipynb).

- Does the given targets match the natural clusters in the data?

  This will be solved in the appended *Notebook*(cluster.ipynb).

- If not, which algorithm is the best at finding the natural clusters (internal evaluation)?

  This will be solved in the appended *Notebook*(cluster.ipynb).

- You should include at least two different algorithms in the comparison, KMeans and Agglomerative.

  This will be implemented in the appended *Notebook*(cluster.ipynb).

- Describe the dataset. What is the purpose of the dataset? What are the features, and what is the target? Write with your own words.

  - **Description of the dataset**
    - These datasets contains a collection of evaluation data on red and white wine variants produced from the Portuguese "*Vino Verde*".
    - The input of data contains objective physicochemical measures of different values in the wines. These are numeric values for variables such as acidity, pH, alcohol and others.
    - The output in the data consist of subjective sensory evaluations on the given wine quality made by wine experts.
    - The quality values are ordered from 0 to 10, whereas 0 is considered poor quality and 10 is considered high quality.
    - Documentation for the datasets states unsertainty around relevance regarding input variables and encourages different feature selection methods to be applied.
    - The datasets contains 1599 instances of red wine and 4898 instances of white wine, with no missing attribute values.
    - Each instance contains 12 attributes, where 11 are input variables and 1 is the output variable (quality).

  - Content of the datasets are provided as ordered and unbalanced. This implies that the data contains a higher volume of *normal* wine qualities than high or low quality wines.
  - According to appended documentation for the datasets, there are no missing values in the data.
  - The datasets are available at the UCI Machine Learning Repository and was donated by Paulo Cortez, University of Minho, Portugal.

- **Purpose of the dataset**
  The purpose of the datasets is to model wine quality based on the given input variables. This can be used to predict the quality of a given wine based on its physicochemical properties.
  According to documentation is considered fitting for classification and regression tasks. In this task, clustering algorithms will be applied to explore the natural clusters in the data and compare these with the given quality targets.
- **Features of the dataset**
  Features of the dataset are the following 11 input variables which all are numeric values:
  1. fixed acidity
  2. volatile acidity
  3. citric acid
  4. residual sugar
  5. chlorides
  6. free sulfur dioxide
  7. total sulfur dioxide
  8. density
  9. pH
  10. sulphates
  11. alcohol
- **Target of the dataset**
  Target of the dataset is the quality variable which is an ordered numeric value ranging from 0 to 10. This variable is a subjective evaluation made by wine experts.

The following should be done for both variants (red,white) of the dataset.

- Explore the dataset using distribution and correlation analysis and follow up with appropriate processing

  This will be solved in the appended *Notebook*(cluster.ipynb).

- Your goal is to find the best possible clustering algorithm, so you must use hyperparameter tuning when possible. The choice of tuning parameters and their values and corresponding results should be discussed. It is recommended that you use manual tuning

  This will be solved in the appended *Notebook*(cluster.ipynb).

- The external evaluation must include the adjusted rand index, homogeneity, completeness, and the v-measure. All results should be discussed

  This will be solved in the appended *Notebook*(cluster.ipynb).

- The internal evaluation must include the Silhouette score and the Davies-Bouldin index. All results should be discussed

  This will be solved in the appended *Notebook*(cluster.ipynb).

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